Expression vectors: An expression vector is a cloning vector containing the regulatory sequences (promoter sequence) necessary to allow the transcription and translation of a cloned gene or genes. Expression vectors are essentially derivatives of the plasmid cloning vectors used in the host. They are used to produce the protein encoded by a cloned gene in the transformed host. For example, the biotechnology industry produces pharmaceutically active proteins with the use of expression vectors and the appropriate host.



A genomic library is a collection of clones that contains at least one copy of every DNA sequence in an organism's genome. Like libraries with books, genomic libraries are a great source of information; in this case, the information is about the genome. Specific sequences in cDNA libraries and genomic libraries can be identified via a number of approaches, including the use of specific antibodies, cDNA probes and oligonucleotide probes

Human artificial chromosome (HAC)-based vectors offer a promising system for delivery and expression of full-length human genes of any size into human cells, and a tool for determining human chromosome function. It does not have the problem of limited cloning capacity of other vectors, and it also avoids possible insertional mutagenesis caused by integration into host chromosomes by viral vector.

12.10 Restriction Enzymes

In 1960s Swiss microbiologist Werner Arber and American microbiologists Hamilthon Othanel Smith and Daniel Nathans discovered restriction enzymes. The discovery, for which the three men shared the 1978 Nobel Prize for Physiology or Medicine. Restriction enzymes or restriction endonucleases are one of the most important groups of enzymes for the manipulation of DNA. It is one of the important molecular tools used by a genetic engineer. These are the bacterial enzymes that recognize a specific base sequence in a DNA molecule (from any source) and make two cuts one in each strand generating 3′ – OH and 5′ – P termini. They were first discovered in E.*coli*. E.*coli* produces the restriction enzyme to cut the viral DNA and protect itself. The host E.coli DNA is protected by its own restriction enzyme since its methylated. Since these enzymes restrict the viral replication the word restriction is added to these enzymes. Hind II was the first discovered restriction endonuclease.

The site where the DNA is cut by a restriction enzyme is called recognition sequence. Restriction endonucleases can specifically recognize DNA with a particular sequence of 4-8 nucleotides and cleave. Each recognition sequence has two fold rotational symmetry i.e. the same nucleotide sequence occurs on both strands of DNA which run in opposite direction. Such sequences are referred to as palindromes, since they read similar in both directions (forwards and backwards). Majority of restriction endonucleases (particularly type II) cut DNA at defined sites within recognition sequence. Type II restriction enzymes make

TN_GOVT_XII_Micro_Biology_CH12.indd 277 04-03-2019 10:51:53